

Feb 6

P₁ ♂ Yellow Seeds YY × ♀ green Seeds yy
↓
100% Yellow Yy

♂ green × ♀ yellow
↓
100% yellow

* الصفة التي تظهر 100% بأول جيل dominant وذاتها التي والتي تظهر 75% في الجيل الثاني.

* The results disproved homunculus law (only one parent inherits features)
* " " " " blending law (2 colors produce another)

P₁ نفس الفوق

F₁ Yellow × yellow
75% 25%

Yellow × Yellow
75% 25%

$\frac{750}{250}$

$\frac{250}{250}$ (اصفر على الاصفر)

yellow
dominant

green
recessive

3

:

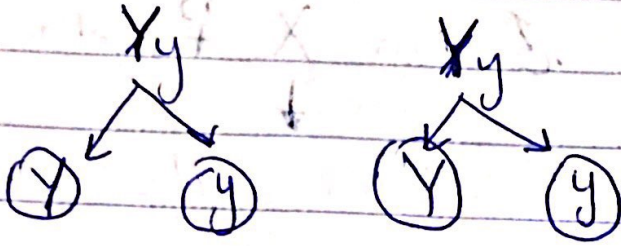
1

yellow

green

F₂ Phenotypic ratio
genotypic ratio.

F₁ Yellow X Yellow



monohybrid

YY, Yy, yY, yy

25%

yellow

homozygotes
dominant

50%

yellow

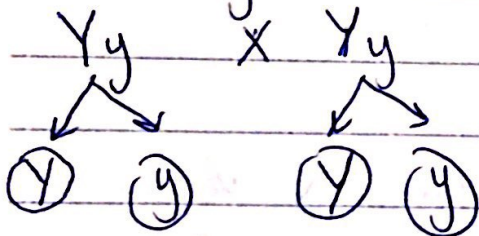
heterozygotes

25%

green

homozygotes
recessive

Probability laws.



→ Law of segregation they will unite again later in fertilization.

$$P(yy) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

(احتمال ياتي من خلال طرف واحد واحدة) (X)

احتمال من الازم

Multiplication rule when traits occurring at same time

$$P(Yy) = P(Yy) \text{ or } P(yY)$$

$$= \frac{1}{4} + \frac{1}{4}$$

$$= \frac{1}{2}$$

* Multiplication & Addition Rule *

Multiplication

* independent events in sequence.
* and

* Flipping 5 coins & getting heads

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

Same as saying having 5 kids.

Addition

mutually ^P exclusive events
or

* Getting heads or tails

$$\frac{1}{2} + \frac{1}{2} = 1$$

In Cross between $AaBbCc \times AaBbCc$

what is the probability that the offspring will be $AA Bb Cc$ or $AA BB CC$

$$\frac{1}{4} \times \frac{1}{2} \times \frac{1}{2}$$

+

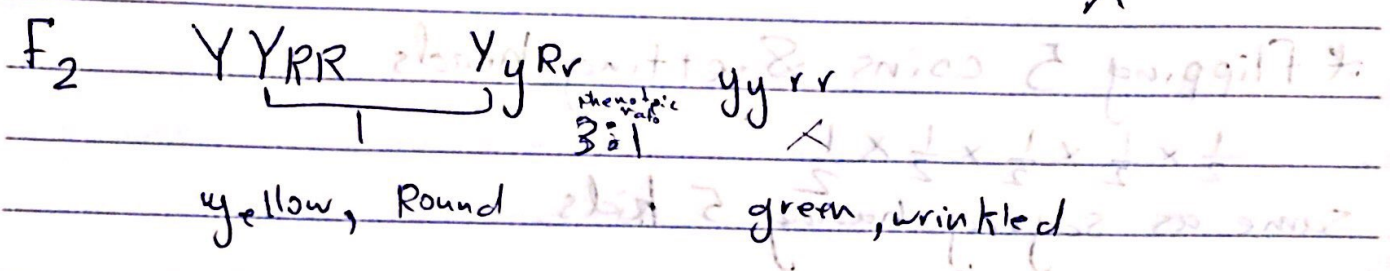
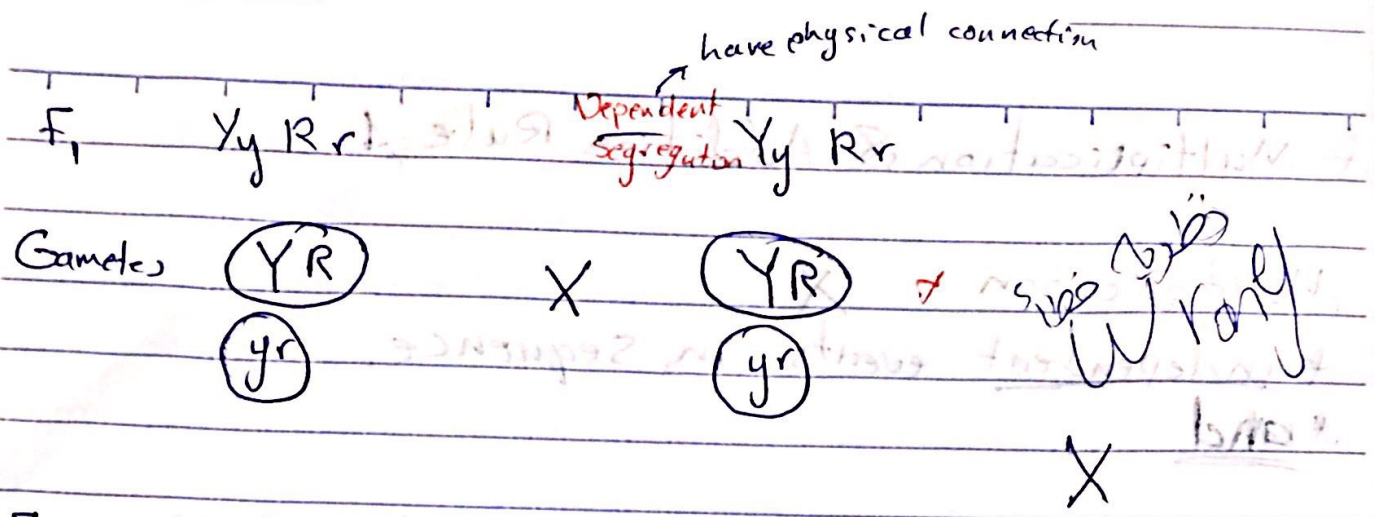
$$\frac{1}{4} \times \frac{1}{2} \times \frac{1}{2}$$

$$\frac{1}{16}$$

+

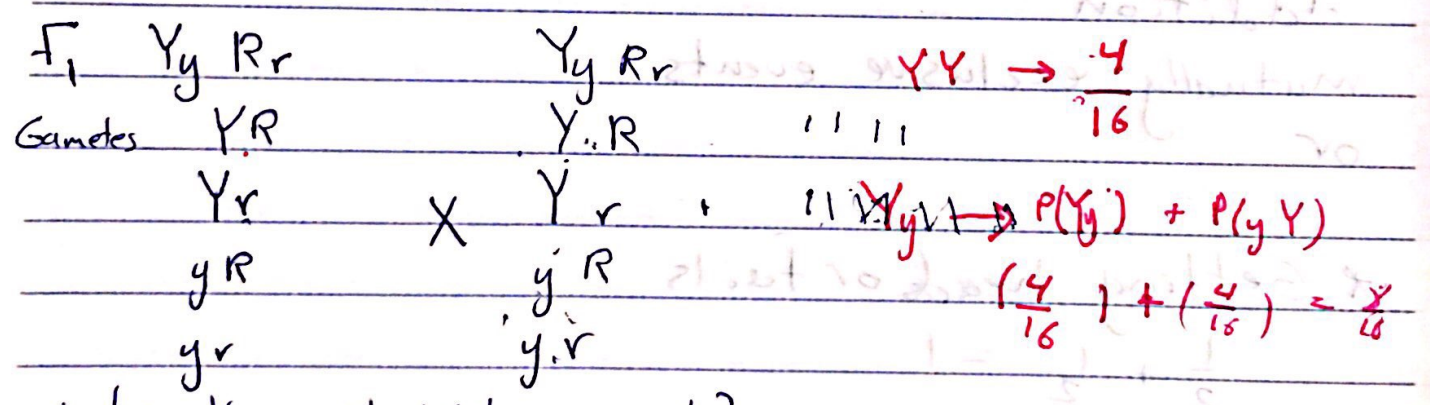
$$\frac{1}{16}$$

$$= \frac{2}{16} = \frac{1}{8}$$

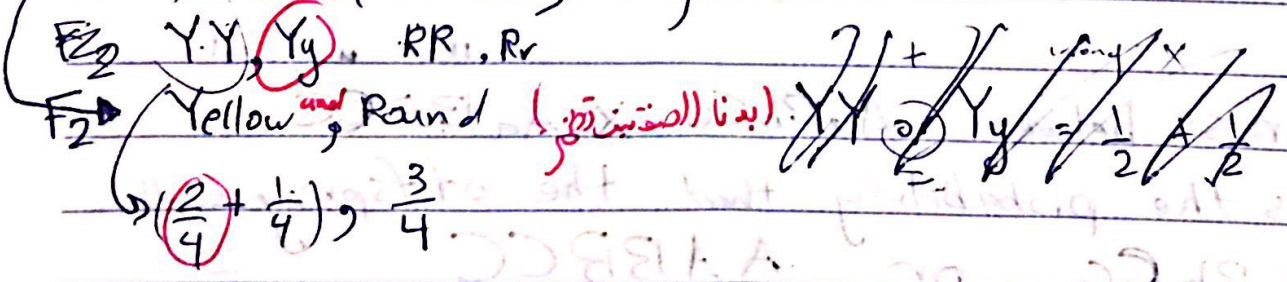


But Mendel didn't get this ratio !!

So the genes aren't dependent !! (independent segregation)



what is the probability to get?



$$\frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$$

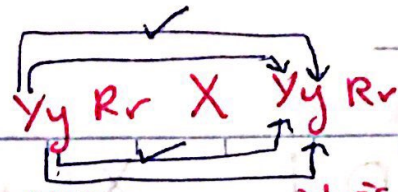
isn't the same as?

$$P(YYRR) + P(YyRR) + P(YyRr) + P(YyRr)$$

$$(\frac{1}{4} \times \frac{1}{4}) + (\frac{1}{4} \times \frac{2}{4}) + (\frac{2}{4} \times \frac{1}{4}) + (\frac{2}{4} \times \frac{2}{4})$$

$$= \frac{1}{16} + \frac{2}{16} + \frac{2}{16} + \frac{4}{16} = \frac{9}{16}$$

ratio 1:3, $\frac{2}{4}$



YY, Yy
yellow and rr
wrinkled

this or this gives yellow wrinkled.
 $P(YYrr) + P(Yyrr)$
 $(\frac{1}{4} \times \frac{1}{4}) + (\frac{2}{4} \times \frac{1}{4}) = \frac{3}{16}$

$$\frac{3}{4} \times \frac{1}{4} = \frac{3}{16}$$

yy RR, Rr
green, Round

$P(yyRR) + P(yyRr)$
 $(\frac{1}{4} \times \frac{1}{4}) + (\frac{1}{4} \times \frac{2}{4}) = \frac{3}{16}$

$$\frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$$

yy rr
green, Round

$P(yyrr)$

$$\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$

$$P(YYRR) = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$

$$P(YyRR) = \frac{2}{4} \times \frac{1}{4} = \frac{2}{16}$$

$$P(YyRr) = \frac{2}{4} \times \frac{2}{4} = \frac{4}{16}$$

X because they are
 Separate
 out come of first
 event separate from
 the other

(independent)

$$Aa Bb Cc Dd \times aa Bb Cc Dd$$

$$P(Aa Bb Cc Dd) = \frac{2}{4} \times \frac{2}{4} \times \frac{1}{4} \times \frac{1}{4}$$

اختلاف ضرب كل واحد

* Test cross:- (to test if one is YY or Yy)

$$P_1 \quad YY \times yy$$

$$F_1 \quad Yy \times Yy$$

$$F_2 \quad YY, Yy, yy$$

75% 25%

3:1 →

Yy يكون heterozygote

① YY × yy

① if 100% yellow → homozygous → homo gen يكون

② Yy × yy

50% YY
50% yy

Cross test
for dihybrid.

F₁ Yy Rr X Yy Rr

F₂ Yy rr X green, wrinkled
Yellow, wrinkled

YY → 100% yellow, wrinkled

50% yellow, wrinkled

50% green, "

F₂ yellow, Round

① YYRR X yyrr

② Yy RR X yyrr

③ Yy Rr X yyrr

YR
Yr
yR
yr

Yy Rr Y, R
Yy rr Y, w
yy Rr G, R
yy rr G, w

F_1 $Yy Rr$

X $Yy Rr$

$1 : 1 : 1 : 1$

$\frac{Y}{y} \quad \frac{r}{R}$

YR, Yr
 yR, yr

Feb 13

CH 3

1 character

2 genes

1 gene

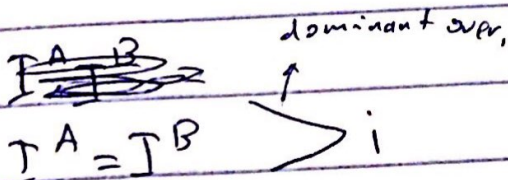
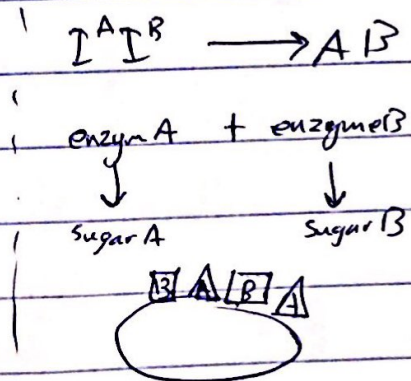
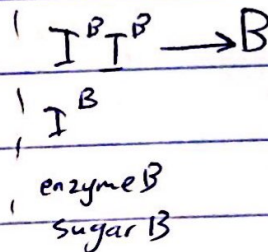
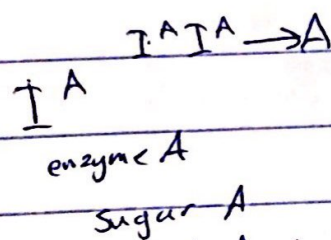
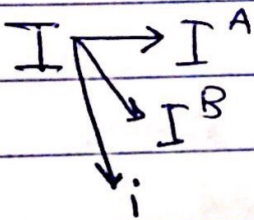
1 gene

multiple alleles

Dominancy relationship

$A > a$

slide 11 Ch. 3



100

- 35 $I^A I^A$
- 15 $I^A i$
- 10 $I^B I^B$
- 10 $I^B i$
- 30 ii

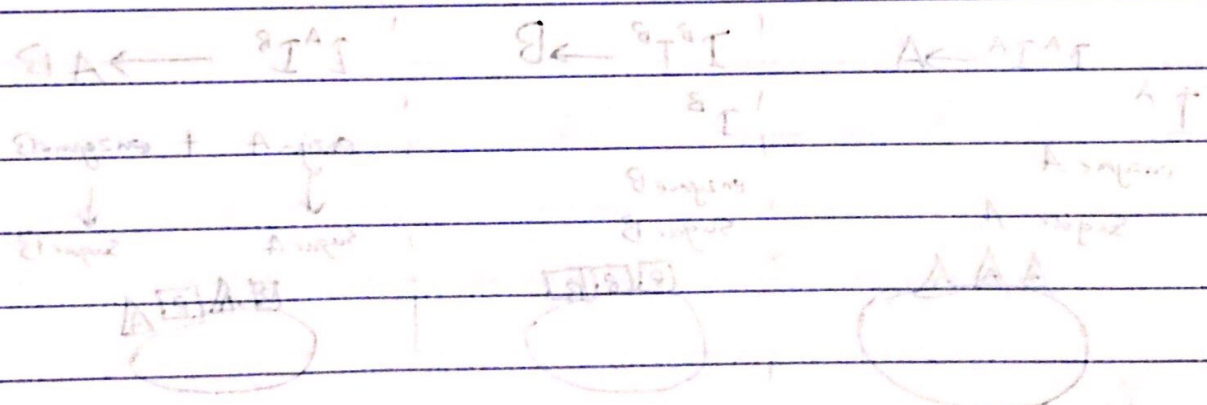
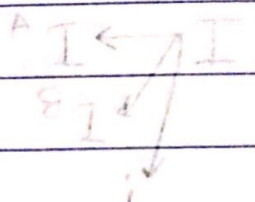
Alleles from both.

$I^A I^A + I^A i$

$$I^A = \frac{(35 \times 2) + 15}{200} = \frac{85}{200}$$

200

All # of Alleles



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Slide 20

Wild type allele :- Most common (more frequent)

Mutant allele :- less common frequency $< 1\%$

Common variants : equal frequencies like blood alleles

Polymorphism $> 1\%$ frequency

CFTR > 6000 alleles

wild type 99%

mutant alleles $< 1\%$

allele 2 5% $\rightarrow$ we can't call it mutant allele,
because freq $< 1\%$

this means $\downarrow$

this allele may
give a function

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Yellow x Yellow

$A^y A$ $A^y A$

2 : 1

← this ratio means there is lethality

$A^y A^y$ (Dead) , $2 A^y A$ (yellow) , AA (Agouti)
66% 33%
2 : 1

Recessive lethality because it needs two alleles

color

$A^y > A$

yellow allele is dominant over agouti

yellow > agouti allele

Lethality

$A > A^y$

A allele is dominant over A^y in case of lethality because $A^y A^y$ doesn't die.

~~Poly~~ Pleiotropy one gene controls many phenotypes

Tan
AA bb

Grey
aa BB



Brown X Brown
Aa Bb Aa Bb

9/16 Brown

$$A- B- \\ \frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$$

or

$$A- = AA + Aa \\ = \frac{1}{4} + \frac{2}{4} \\ = \frac{3}{4}$$

3/16 tan

$$A- bb \\ \frac{3}{4} \times \frac{1}{4} = \frac{3}{16}$$

3/16 gray

aa B-

1/16 green

aa bb

Tan AA or Aa bb

Grey BB or Bb aa

Brown ~~Aa Bb~~ A- B-
~~AA BB~~

slide 33

 F_1 $Aa Bb \times Aa Bb$
 F_2 ① $A-B-$ $\rightarrow$ Brown

 ② $aa B-$ $\rightarrow$ gray

 ③ $A-bb$ $\rightarrow$ Tan

 ④ $aa bb$ $\rightarrow$ green
white $\times$ white $AA bb \times aa BB$ 
 Purple $\times$ Purple
 $Aa Bb \times Aa Bb$

 2 genes are important
 for one trait.

 $9/16$ $A-B-$
 $7/16$ $A-bb$
 $aa B-$
 $aa bb$

 (Pastly was 2 gene
 for 2 different traits)

Black X yellow

BBEE

bbee

epistatic gene.

B- E-



ee applies regardless of b ph

recessive epistatic

Black

X

Black

BbEe

BbEe

9/16



Black

B- E-

3/16



Brown

bbE-

4/16



yellow

B- ee 3/16

bbee + 1/16

♂

○

ii

♀

○

ii

↓

ii

Appears to be ○

♂

○

x

♀

○

$I^A I^A; hh$

$ii; Hh$

True ○

~~$I^A I^A; hh$~~ (ii) I^A → $I^A i; Hh$ → A



○

$ii; H-$ / $ii; hh$

$- -; hh$

A : $I^A I^A; H-$
 $I^A i; H-$

Epistasis is recessive
Need 2 genes to mask

$H-$ epistatic

ii hypostatic

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SI

White X green



white X white

12/16 → white A-B-

aaB-

3/16 → yellow

A-bb

one allele of A

نظر وجود bb (قوي يسبح قوي الصفة)

1/16 → green

aabb.

Slide 45

P White X white

AA BB

aa bb

$\frac{F_1}{4}$

F₁ white X white

Aa Bb

Aa Bb



F₂ 13/16 → white A-B-

aa B-

aabb

1

3/16

Brown

A-bb

$AaBb \times AaBb$

AB

Ab

aB

ab

f_2	$A-B-$	$9/16$	$\rightarrow$ Brown
	$aaB-$	$3/16$	$\rightarrow$ tan
	$Aabb$	$3/16$	$\rightarrow$ grey
	$aabb$	$1/16$	$\rightarrow$ green

Heterogeneous $\rightarrow$ complementary testing

~ 50 100 genes $\rightarrow$ Deafness

mutation in any of those gene caus deafness

How can we know if mutation is from same gene or not, by Complementation test ^{in different person}

σ	$\times$	ϕ	
Deaf		Deaf	
$aaBB$		$AAbb$	
$\downarrow$		$\downarrow$	
Normal		100%	
$AaBb$			
mutation from 2 different genes		complemente	

Deaf	$\times$	Deaf
$aaBB$		$aaBB$
$\downarrow$		$\downarrow$
100% Deaf		
fail to complement		

13/1/19

① hypothesis

①

Albino bb x Brown BB

↓

Black Bb x Black Bb

- ~ 25% Albino bb
- ~ 50% Black Bb
- ~ 25% Brown BB

Independent

hypothesis

فرضیه مستقل

F_2 Albino bb x Brown BB

↓

100% Black.

② hypothesis

recessive to c

Albino $BbCc$ x Brown $bbCC$

F_1 ~~$BbCc$~~ $BbCc \rightarrow$ Black

↓

$a:3:4$
 $t:2:1$
 Albino
 by dividing by 16 they are very close
 Brown 1
 $bbc- B-c- 13-cc$
 $bbcc$

Test hypothesis

♂

Albino x ~~brown~~

$bbcc$ x $bbCC$

↓

Brown $bbCc$

♀

$BbCc$ x $bbCC$

↓

$BbCc$ 100% Black.

$RB^+ (RB)$ ^{mutant}

Individual with same genotype
by have different phenotype

$RB^+ RB^+$ $\rightarrow$ 70% Reticoblastoma
 $\rightarrow$ 30% Normal

70% Penetrance.

~~$AaBbCc \times AaBbCc$~~

$A'A^0 B'B^0 C'C^0 \times A'A^0 B'B^0 C'C^0$

dark
 $P(A'A^0 B'B^0 C'C^0) = \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{1}{64}$
 $\rightarrow$ degree 6

light
 $P(A^0 A^0 B'B^0 C^0 C^0) = \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{1}{64}$
 $\rightarrow$ degree 0

$P(A'A^0 B^0 B^0 C^0 C^0) = \frac{1}{2} \times \frac{1}{4} \times \frac{1}{4} = \frac{2}{64}$
 $\rightarrow$ Degree 1

$P(A^0 A^0 B^+ B^0 C^0 C^0) = \frac{2}{64}$
 $\quad \quad \quad + \quad \quad \quad c^+ c^0$

A^+ : dark
 a^0 : light.
 B^+ : dark
 b^0 : light
 C^+ : dark
 C^0 : light.

7 Phenotypes $\rightarrow$ 0 to 6

1-5%

Feb 27

Second

Chapter 4

Gp 6 islands $\rightarrow$ C^{meth}

① Promoter

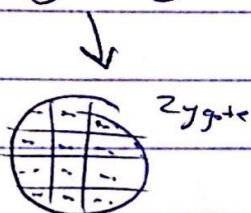
$\rightarrow$ mRNA

② Histone modification $\rightarrow$ affect gene expression

Maternal inheritance.

Zygotic genotype (by both Parents)

♀ Egg ♂ Sperm



$\rightarrow$ traits affected by the mother's genotype indep. of the zygotic genotype

دائري

D: dextral

d: sinistral

$D > d$

mother

♀ Sinistral X ♂ Dextral

dd DD

Sinistral Dd

(because of mother)

♀ Dd X ♂ dd

Dextral (D > d) cause

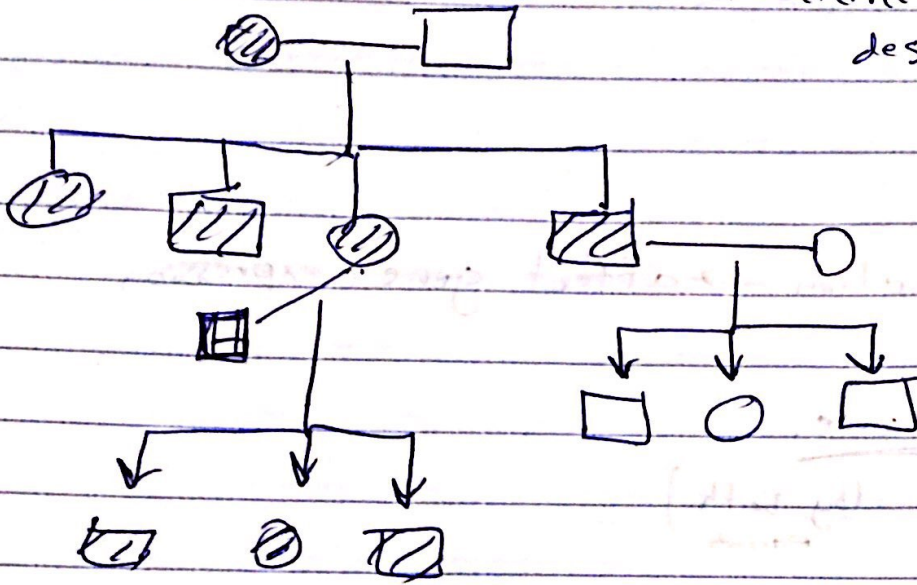
♀ Dextral DD X ♂ Sinistral dd

Dextral ♀ Dd X dd

$\rightarrow$ dextral

mother's genotype is more cross is

Maternal inheritance
disease



Qs data show

if autosomal dominant
one

AA Aa

one parent affected

X
doesn't work

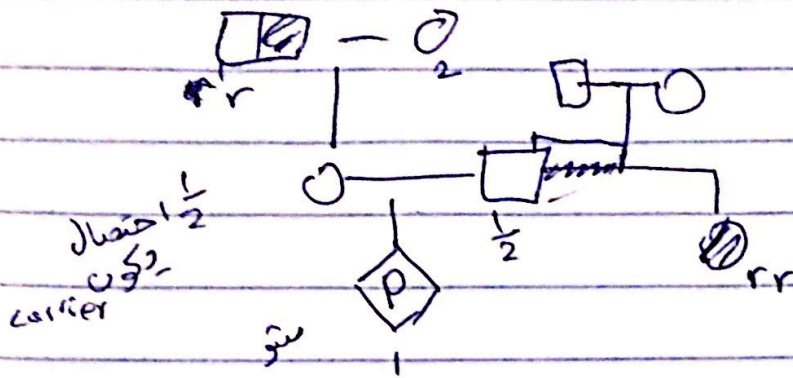
III-1 → Aa

II-2 → AA or Aa

because he has a sister like

III-5 → Aa or AA

RR or Rr



Mother carrier x Father carrier x Son c

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{4} = \frac{1}{16} \leftarrow \text{affected}$$

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8} \leftarrow \text{carrier}$$

2 characteristics →

homozygous → meaningful breeding

$$F K^+ F K^+ g l s^+ g l s^+ \times f k f k g l s g l s$$



wild type wild

$$F_2 \quad F K^+ - g l s^+ \quad \frac{9}{16}$$

$$F_2 \quad K k f k \quad g l s^+ \quad \frac{3}{16}$$

this is phenotypic ratio

genotypic ratio

$$F K^+ F K^+ g l s^+ g l s^+ \rightarrow \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$

$$F K^+ f k g l s^+ g l s^+ \rightarrow \frac{1}{2} \times \frac{1}{4}$$